

Privacy-Preserving Employee Database Using Partial Homomorphic Encryption and Searchable Encryption

Project Goal

This assignment introduces students to practical applications of:

- Partially Homomorphic Encryption (PHE)
- Searchable Encryption (SE)
- Secure Outsourced Databases
- Privacy-Preserving Query Processing

Students will build a secure employee database system using the provided dataset that must be capable of performing selected computations and searches over encrypted data."

1. Provided Dataset

The provided dataset contains employee information with typical data as found in a company employee database. You must analyze the dataset to investigate the data.

Suppose the company wants to outsource its employee database to an untrusted cloud provider (supported by a remote database). The cloud provider:

- must NOT know and not learn about any plaintext employee information
- must still support selected queries and computations required to process data in the database, maintaining the database always encrypted

2. Architecture for the Solution

Students must design and implement a secure system where the given dataset must be supported, with all information encrypted on a remote server, with protection against an "Honest-But-Curious" System Administrator. Students are free to select the data management system to store the dataset on the server side. No cryptographic keys must be known by the server.

- Can use an SQL Database (e.g., MySQL or any other)
- Can use a NoSQL Database (MongoDB, CosmodDB, etc.)
- Another alternative is to use Filesystem

Each group/student may and should use the technology they can best leverage considering prior knowledge and practical experience

The solution must support the following operations (in section 4) which must be processed by the server directly over the encrypted data. These operations are

requested by the client side. Only the client must have the keys and know how to perform the operations, according to the cryptographic methods and algorithms used.

3. Implementation Language

Java is the primary implementation language (for immediate reusing of initial demos/examples/discussions from Labs).

Other programming languages (e.g., Python, Go) may also be used. However, choosing these languages requires students to research implementations or cryptographic libraries for all cryptographic algorithms that must be used (as discussed below).

4. Operations that must be supported

4.1 Client Bootstrap

A client bootstrap must be implemented to submit the encrypted database and to build an index with all required keys or cryptographic parameterizations needed to support the different operations to be performed by users and their client applications."

4.2 Searchable Encrypted and Homomorphic Encrypted Operations on Server Encrypted Data (Server Database)

The operations must be implemented in client application to access and perform the following operations on the remote server:

- Search and retrieve registered information by Employee Identification
- Search and retrieve registered information by Employee Full Name
- Search and retrieve Employees by an ordered list of their salaries
- Search and retrieve Employees belonging to a given Department
- Search and retrieve the Employee with the highest salary
- Compare whether an employee has a higher salary than another, given the Full Names of both employees
- Search and retrieve the list of employees ordered by their ages
- Obtain the salary of a given list of employees converted to US Dollars (using an illustrative Euro–US Dollar exchange rate)
- Find all employees in a given Department AND compute the encrypted payroll sum of their salaries to obtain the total payroll of the list
- Find the oldest employee and retrieve all registered information for that employee
- For all employees with Bonus Eligibility, compute the bonus to be awarded according to the company's bonus policy: 25% of the registered salary

All operations must be processed by the server directly on stored encrypted data. The server will not have any cryptographic keys and will not know the algorithms used (in bootstrap) to encrypt the data maintained in the database. To facilitate the implementation, students may implement separate test client code to support each operation (in different programs). All test clients will share the index obtained in the bootstrap operation.

5. Cryptographic References

References for searchable encryption and partial homomorphic encryption algorithms are provided below. Students must understand how the algorithms are to be used to support the required operations.

- Deterministic symmetric encryption for exact-match queries (AES/CTR), using a 256-bit key
- Secure hash functions for indexing exact-match queries, using SHA-256
- Paillier Cryptosystem for operations requiring additive homomorphic encryption, with a 2048-bit key
- Unpadded RSA with a 2048-bit key for operations requiring multiplicative homomorphic encryption
- ElGamal with a 2048-bit key for operations requiring multiplicative homomorphic encryption
- Boldyreva algorithm for Order-Preserving Encryption, using a 256-bit security level for the underlying Pseudo-Random Function (based on HMAC-SHA256) or AES-256 symmetric block cipher
- ECDSA for attaching a digital signature for authentication of each entry in the database, as a guarantee of authenticity during bootstrap
- HMAC-SHA256 for integrity verification of each database entry, to provide integrity guarantees to clients obtaining the results of the required operations"

6. Project Deliverable

Students must submit the following materials via a Submission Form:

- Source Code: in a GitHub Repository, which must be shared with "henriquejoaolopesdomingos" by the submission date
- A short technical report (in PDF) included in the repository, where the following must be clearly addressed:
 - System Design Assumptions
 - Architecture
 - Database Encryption Choices

- The bootstrap process and the generation of client-side indexed information required to perform the supported operations
- Security Analysis
 - Leakage discussion on the provided solution, clearly stating what the server learns from the operations processed on encrypted data
 - Possible solutions designed and implemented to minimize leakage risk
- Performance Evaluation (from supported instrumentation) in Runtime Demo/Evaluation
 - Latency observed by the client during bootstrap for the encrypted database
 - Encryption time of each algorithm used in encrypting the database attributes
 - Latency (request/response) for each supported client operation ▪ Storage space overhead of the encrypted dataset compared with the original plaintext dataset

7. Important Dates for Project Assignment Delivery

Submission Form and all required delivery elements:

Deadline: 14 June 2026, 23:59

The form for Project Assignment 2 delivery is planned to open after 5 June 2026

8. Reference evaluation criteria

Criteria	Weight	Obs
Architectural Design and Implementation of the system overall	20%	
Correct encryption design to support all required operations, correct implementation of the Bootstrap for the Encrypted Database, and correct generation of all the indexing elements that must be used to support client ioperations	15%	

All Supported Operations and Operations Processing on Encrypted Data (on the Server Side)	35%	Equally divided by all 11 operations correctly supported and implemented for the required client-requests
Report with the requested report information	15%	Appreciation of report, according to reported elements and experimental analysis on implemented operations implemented
Additional highlights: optimizations, code structure and quality, with particular reference to factorization of all cryptographic methods provided in a designed and implemented cryptographic library in the project	15%	Beyond the source code, the use of a jar library in final code supported by client-operations will be appreciated.